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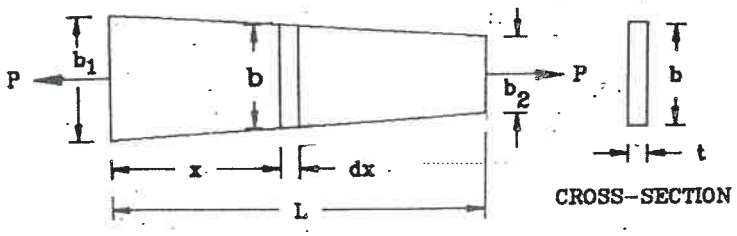
FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

STRENGTH OF MATERIALS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 x02=10**		
1.	**a.**	Define bulk modulus.	**L1**	**1**	**02**
	b.	Discuss the difference between concentrated and distributed loads acting on a beam, providing examples of each.	**L1**	**2**	**02**
	c.	Define pure bending and explain why it typically occurs in symmetric members subjected to uniform loading.	**L1**	**3**	**02**
	d.	Define torsion and explain how it differs from other types of mechanical loading such as tension and compression.	**L1**	**4**	**02**
	e.	Define the transformation of stress and strain in materials and explain its significance in analyzing complex loading conditions.	**L1**	**5**	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Draw the stress strain diagram for ductile and brittle materials with usual notations.	**L2**	**1**	**05**
		OR			
	b.	A bar of uniform thickness 't' tapers uniformly from a width of b1, at one end to b2, at other end in a length 'L' as shown in Fig. 2 (b). Find the expression for the change in length of the bar when subjected to an axial force P.  Fig.2b	**L3**	**1**	**05**
3.	**a.**	A circular rod of diameter 20 mm and 500 mm long is subjected to a tensile force of 45 kN. The modulus of elasticity for steel may be taken as 200 kN/mm^2. Find stress, strain, and elongation of the bar due to applied load.	**L3**	**1**	**05**
		OR			
	b.	A compound bar consists of a circular rod of steel of diameter 20 mm rigidly fitted into a copper tube of internal diameter 20 mm and thickness 5 mm as shown in fig. 3.b If the bar is subjected to a load of 100 kN, find the stresses developed in the two materials. Take E_{steel}= 2x 10^5 N/mm^2 and E_{Copper}= 1.2x 10^5 N/mm^2	**L3**	**1**	**05**

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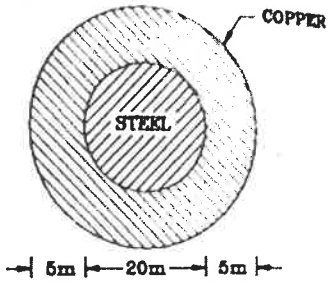


Fig.3b

4.	a.	Derive the relationship between load intensity, shear force and bending moment.	L3	2	05
		OR			
	b.	Draw the bending moment and shear force diagram for the cantilever beam.	L3	2	05

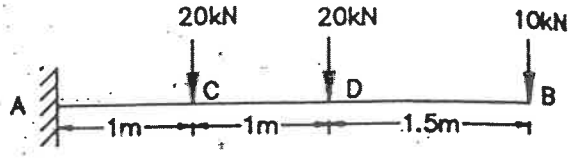


Fig.4b

5.	a.	Explicate the assumptions in simple theory of bending.	L2	3	05
		OR			
	b.	A circular pipe of external diameter 70 mm and thickness 8 mm is used as a simply supported beam over an effective span 2.5 m. Find the maximum concentrated load that can be applied at the centre of the span if permissible stress in tube is 150 N/mm².	L3	3	05
6.	a.	Explicate the assumptions in simple theory of pure torsion.	L2	4	05
		OR			
	b.	A hollow circular shaft 200 mm external diameter and thickness of metal 25 mm is transmitting power at 200 rpm. The angle of twist over a length of 2 m was found to be 0.5 degrees. Calculate the power transmitted and the maximum shear stress induced in the section. Take modulus of rigidity of material as 84 kN/mm^2	L3	4	05
7.	a.	What is stress transformation and why is it important in engineering?	L2	5	05
		OR			
	b.	A circular bar of diameter 25 mm is subjected to an axial force of 20 kN as shown in fig. 7.b Find the stresses on a plane making 30° to the plane of axial stresses and also on the plane which has maximum shear stress.	L3	6	05

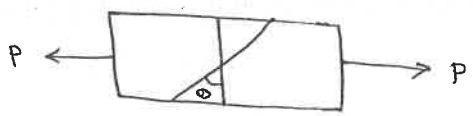
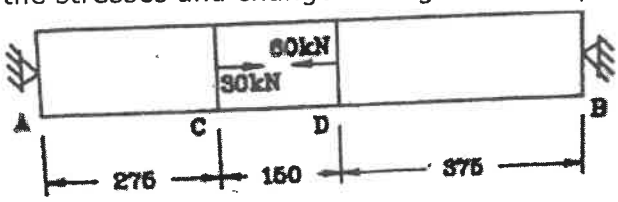
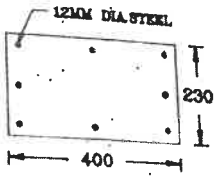
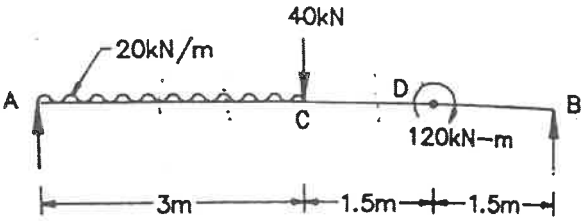
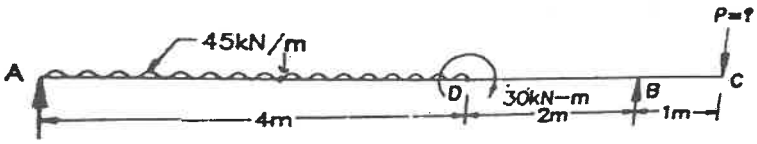
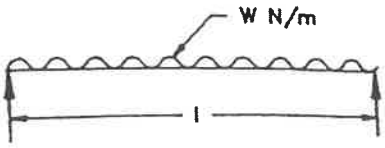
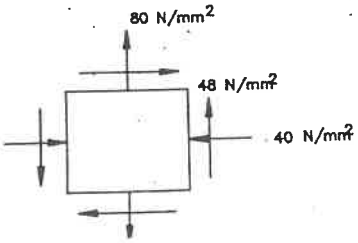


Fig.7b

		PART-C			06 X 10=60
8.	a.	A bar of 800 mm length is attached rigidly at A and B as shown in fig. 8.a. Forces of 30 kN and 60 kN act as shown on the bar. If E= 200 MPa, determine the reactions at the two ends. If the bar diameter is 25 mm, find the stresses and change in length of each portion.  Fig.8a	L3	1	10
		OR			
	b.	A reinforced concrete column of size 230 mm x 400 mm has 8 steel bars of 12 mm diameter as shown in Fig. 8 (b). If the column is subjected to an axial compression of 600 kN, find the stresses developed in steel and concrete. Take Modular ratio= E_s/E_c=18.67  Fig.8b	L3	1	10
9.	a.	A bar of 20 mm diameter is tested in tension. It is observed that when a load of 37.7 kN is applied the extension measured over a gauge length of 200 mm is 0.12 mm and contraction in diameter is 0.0036 mm. Find Poisson's ratio and elastic constants E, G,K	L3	1	10
		OR			
	b.	Derive the relationship between modulus of elasticity and bulk modulus.	L3	1	10
10.	a.	Draw shear force and bending moment diagram for the beam shown in fig 10.a.  Fig.10a	L3	2	10
		OR			
	b.	Draw the SF and BM diagrams for the beam, Mark the salient points and their values on the diagram. Locate the point of contraflexure, if any.  Fig.10b	L3	2	10

11.	**a.**	Derive the relationship between moment and radius of curvature.	L3	3	**10**
		OR			
	b.	A simply supported beam of span 5 m has a cross section 150 mm x 250 mm. If the permissible stress is 10N/mm². find (a) maximum intensity of uniformly distributed load it can carry. (b) maximum concentrated load P applied at 2 m from one end it can carry.  Fig.11a	L3	3	**10**
12.	**a.**	What percentage of strength of a solid circular steel shaft 100 mm diameter lost by boring 50 mm axial hole in it. Compare the strength and weight ratio of the two cases.	L3	4	**10**
		OR			
	b.	A Solid shaft transmits 250 kW at 100 rpm. If the shear stress is not to exceed 75 N/mm², what should be the diameter of the shaft? If this shaft is to be replaced by a hallow one whose internal diameter is 0.6 times outer diameter, determine the size and the percentage saving in wright, the maximum shearing stress being the same.	L3	4	**10**
13.	**a.**	At a point in a strained material there is tensile stress of 80 N/mm² upon a horizontal plane and a compressive stress of 40 N/mm² upon a vertical plane. There is also a shear stress of 48 N/mm² upon each of these planes. Determine the planes of maximum shear stress at the point. Determine also the resultant stress on the planes of maximum shear stress.  Fig.13a	L3	6	**10**
		OR			
	b.	Construct Mohr's circle for a plane stress state with given stress components. Explain in detail how Mohr's circle aids in determining principal stresses and maximum shear stresses. Discuss the practical implications of Mohr's circle in material failure analysis and stress concentration.	L3	5	**10**